

Ideological polarisation—actual divergence of opinions or ‘merely’ a perception effect?

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SSC 2024 in Krakow

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in collaboration with

- Agostino Merico (ZMT; University Bremen)
- Paul E. Smaldino (UC Merced; Santa Fe Institute)

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→ **CSH Vienna**

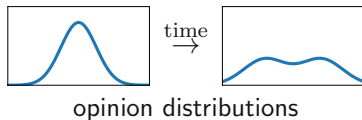
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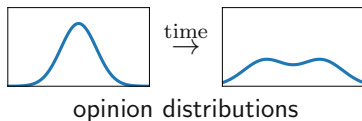
Ideological polarisation and opinion perception

- Ideological polarisation = disagreement $\uparrow$

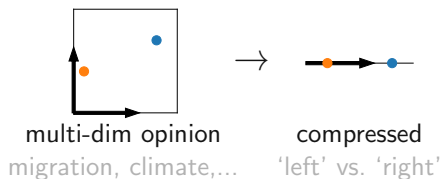


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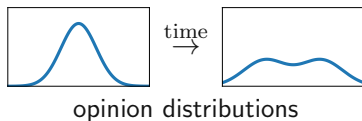


- How do people perceive opinions?

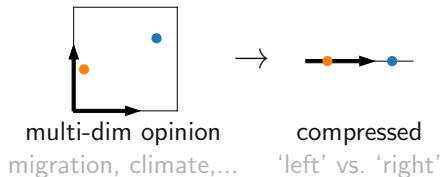


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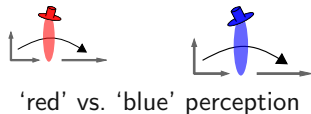
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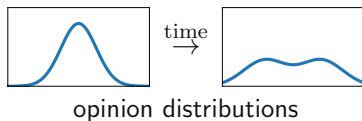


- What shapes people's perceptions?
Here: political identity

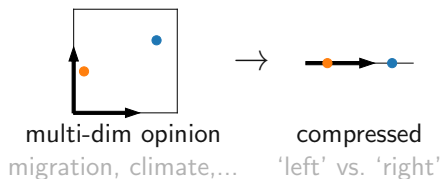


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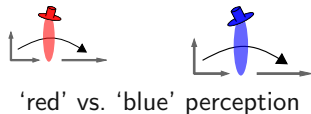
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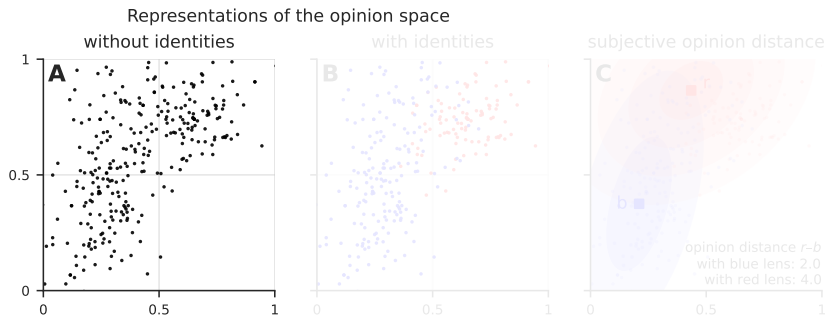


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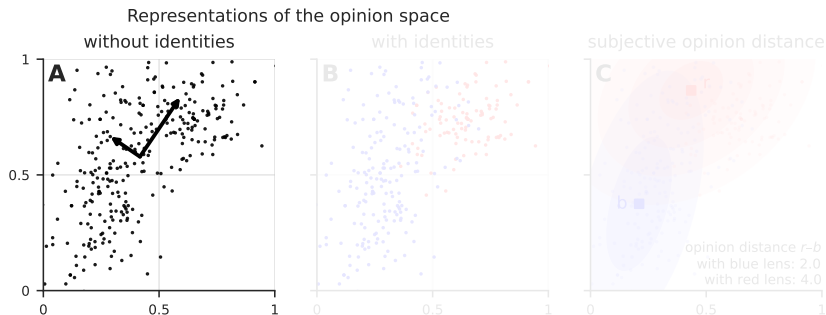
RQ: How do changes in the way people perceive opinions modify the way they see ideological polarisation?

Method: illustrative example



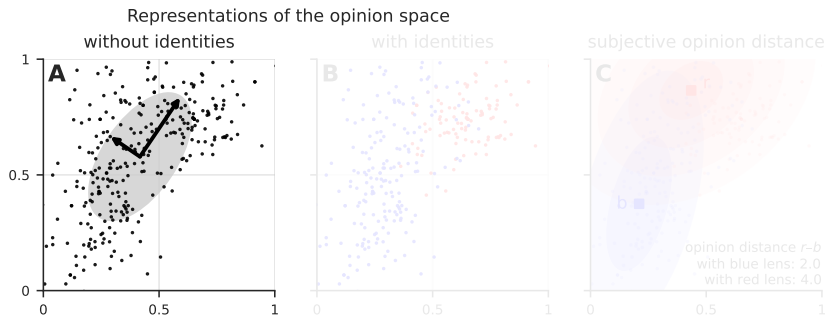
- Principal component (PC) analysis to infer perceived opinion space
→ arrows = $PC \times \text{loading}$

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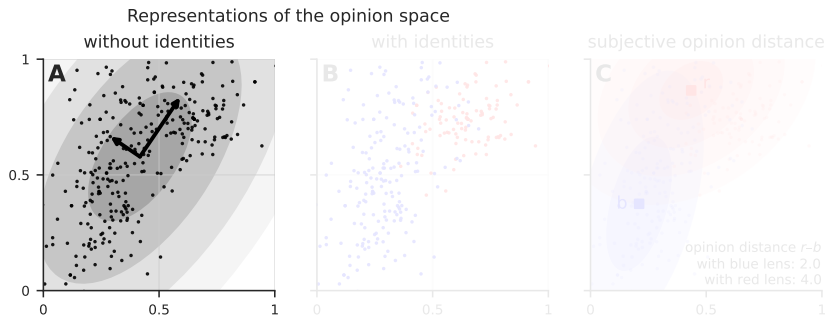
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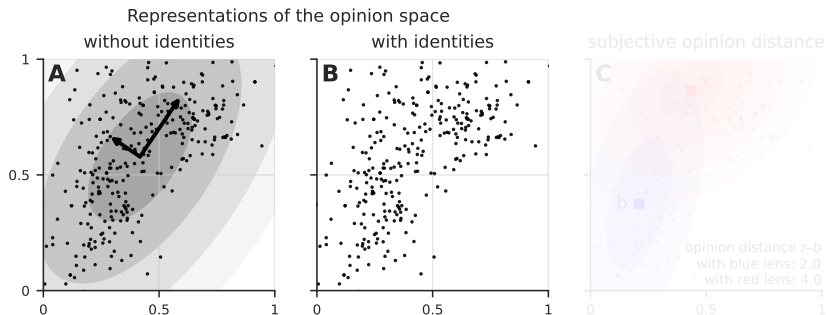
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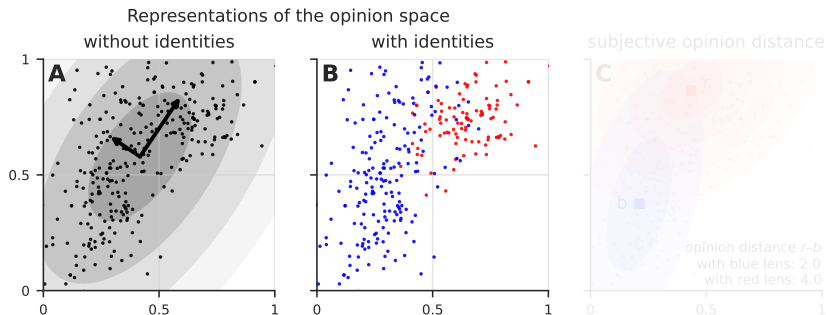
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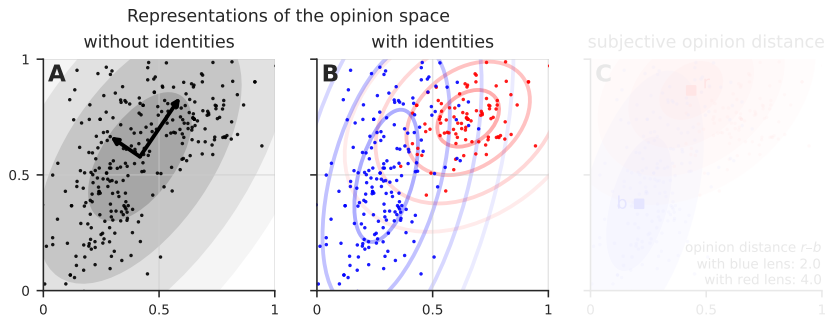
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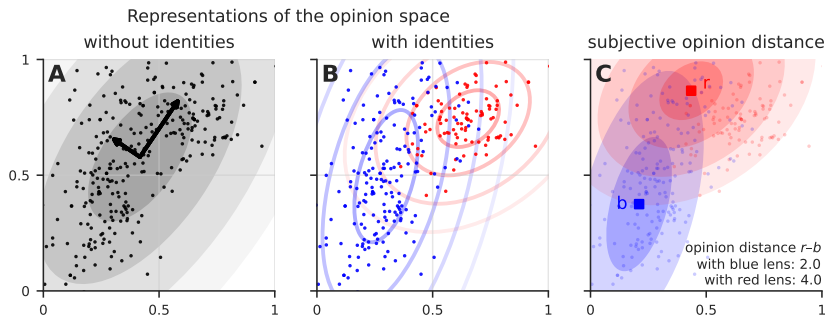
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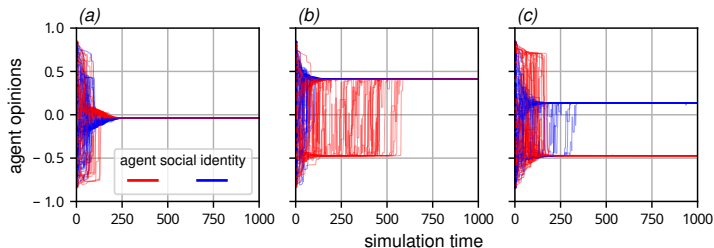
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- If in-group variance low (= in-group homogeneity) →
→ low tolerance → opinion differences over-emphasised.

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→ low tolerance → opinion differences over-emphasised.
- Asymmetric perception of opinion differences.

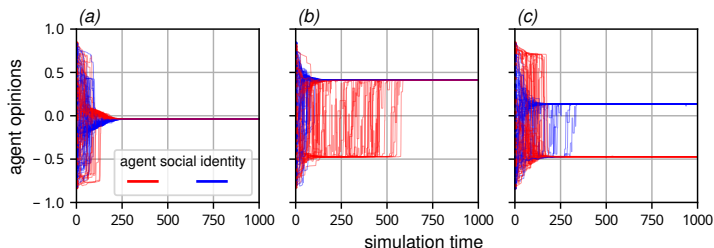
Why at social simulation conference?



Example simulations from an opinion dynamics model in Steiglechner, Smaldino, Moser, and Merico (2023) “Social Identity Bias and Communication Network Clustering Interact to Shape Patterns of Opinion Dynamics” in *Journal of the Royal Society Interface*.

- Opinion dynamics models often involve high-frequency, large opinion jumps & produce very stable patterns in the long run.

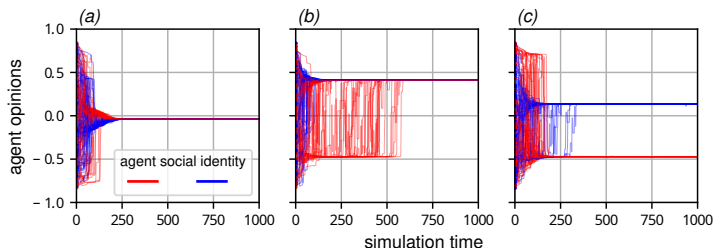
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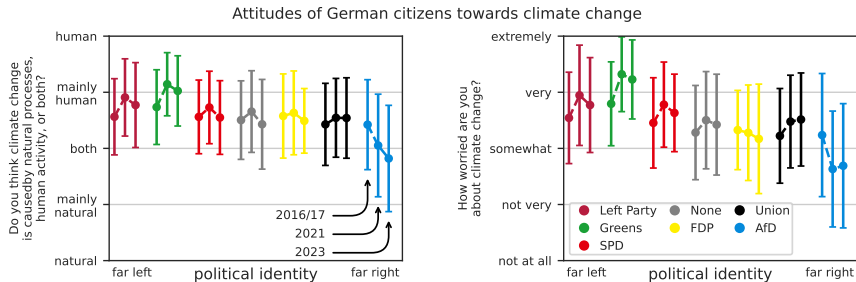
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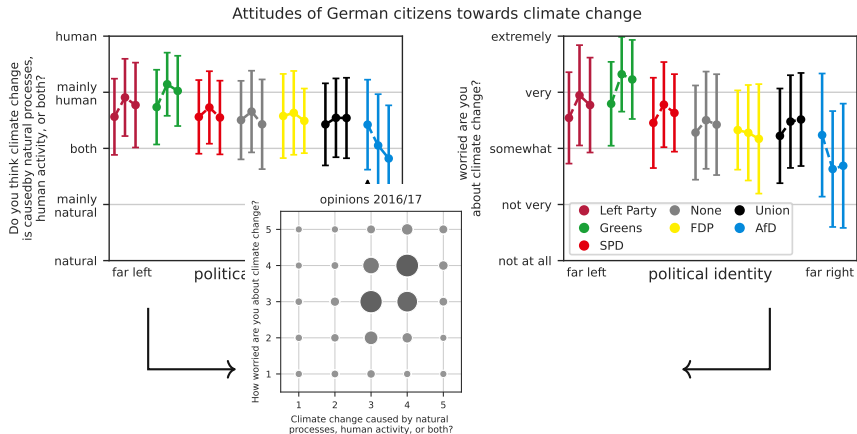
- Opinion dynamics models often involve high-frequency, large opinion jumps & produce very stable patterns in the long run.
- The models focus on divergence of opinions, but omit perception of opinions → are they missing a crucial aspect?
- There is ample empirical evidence for misperceived polarisation (Enders and Armaly, 2019; Levendusky and Malhotra, 2016; Westfall et al., 2015)

Proof of concept: Climate change opinions of Germans



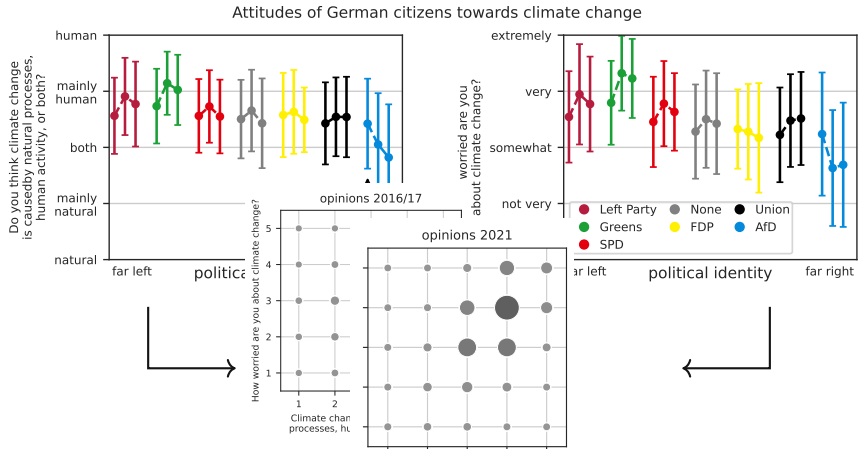
Data taken from the European Social Survey (ESS) in 2016/17, 2021, 2023.

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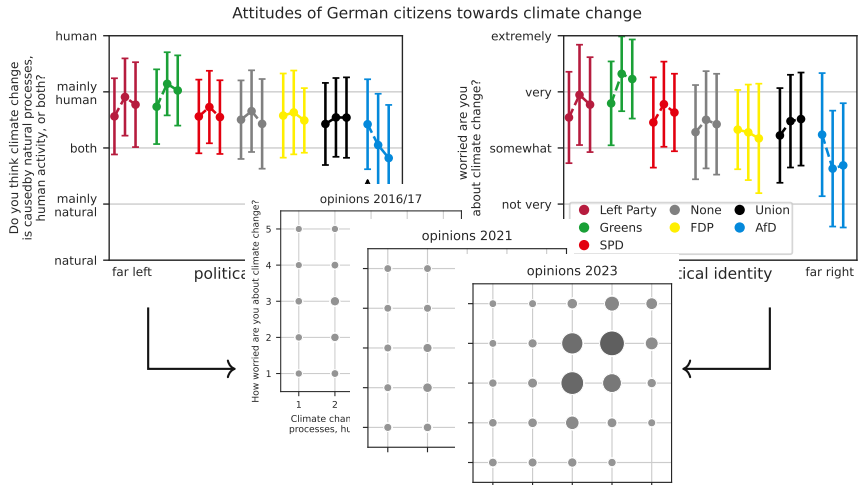
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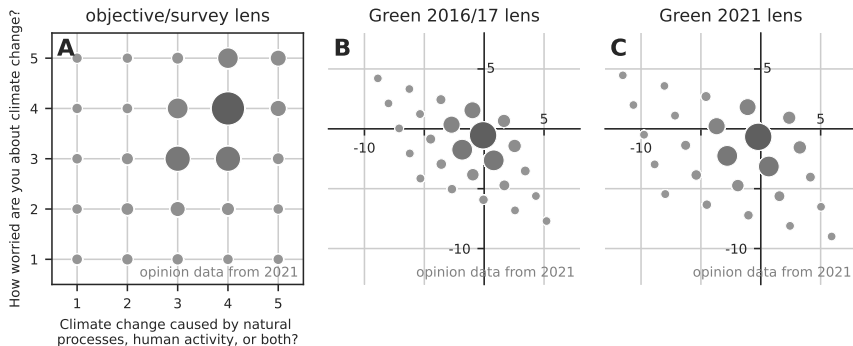
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‘Put yourself in their shoes’— opinion data from a subjective perspective

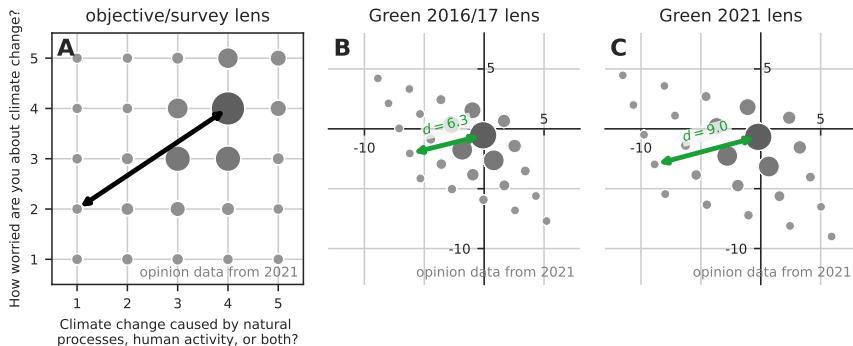
Opinion distribution in 2021 seen from different perspectives:



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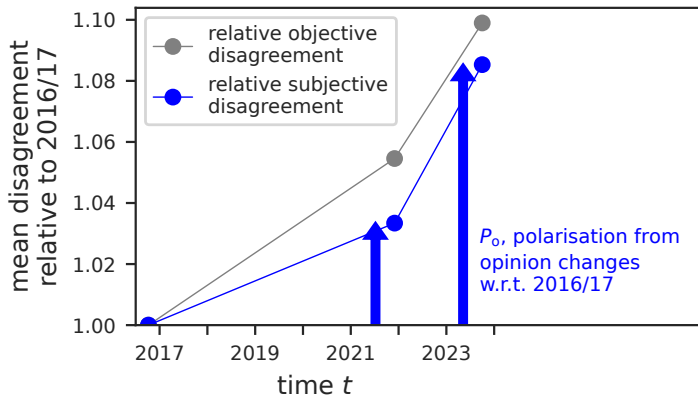
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How do changes in the way people perceive opinions modify the way they see ideological polarisation?

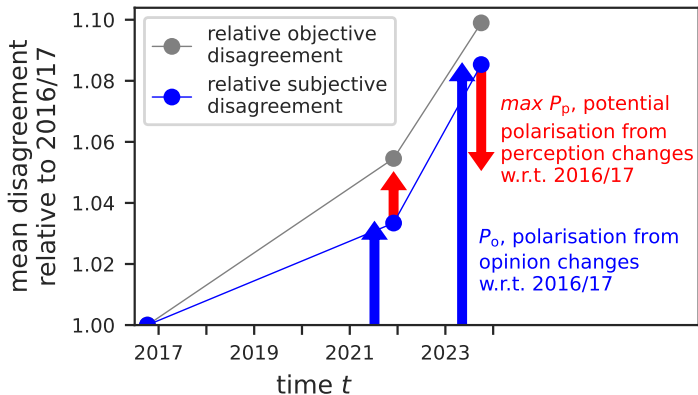
→ ...



* assuming that perceptions are equal to those from 2016/17 and remain fixed thereafter

How do changes in the way people perceive opinions modify the way they see ideological polarisation?

→ **it varies over time ...**



* assuming that perceptions co-evolve and adapt instantaneously to the current in-group opinions

How do changes in the way people perceive opinions modify the way they see ideological polarisation?

→ **it varies over time and across groups!**

Table: $\max P_p$, potential relative polarisation only due to perception changes.

	All	Left	Greens	Social Demo- crats	None	Liberals	Conser- vatives	Right Extrem- ists
	–	Die Linke	Bündnis 90/Die Grünen	SPD	–	FDP	CDU/ CSU	AfD
16/17 → 21	+0.02	–0.02	+0.15	+0.04	–0.01	–0.09	+0.06	–0.02
16/17 → 23	–0.03	–0.10	+0.06	+0.06	–0.08	–0.05	+0.03	–0.12

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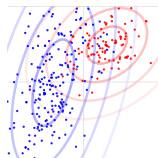
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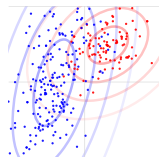
Concluding remarks

- Our hypothesis: opinion perception is driven by in-group opinions and adapts dynamically

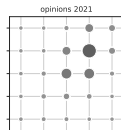


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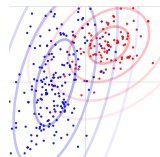


- Applied to the German climate debate:
 - Opinions have diverged to some extent
 - But this changes substantially, *if* the way people perceive opinions co-evolves over time.

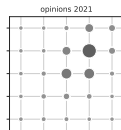


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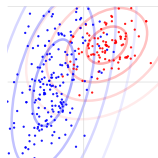


- Our method of looking at data aims to explain why people sometimes see more polarisation than opinion divergence suggests

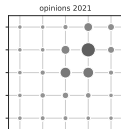


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- Our method of looking at data aims to explain why people sometimes see more polarisation than opinion divergence suggests
- Independent from affective polarisation
(→ additional driver for perception changes?)



Appendix

Notes of caution

- **No network:** We assume that individuals see the opinion distances between themselves and everyone else.
- **No antagonism:** We assume that individuals care about proximity with their in-group, not about distinction from out-groups.
- **No delay:** We assume perceptions adapt immediately to opinion change, although we should expect some lag or inertia here. But our framework would allow such delays.

Method – Equations

Disagreement at time t

$$d(t) := d(\mathcal{X}_t, \mathcal{T}_t) = \frac{1}{n} \cdot \sum_i \frac{1}{n-1} \cdot \sum_{j \neq i} \delta(x_i, x_j | T_i) . \quad (1)$$

with pairwise (subjective) opinion distances between individuals i and j seen by individual i

$$\delta(x_i, x_j | T_i)^2 = (x_i - x_j)^T \cdot (T_i)^{-T} (T_i)^{-1} \cdot (x_i - x_j) . \quad (2)$$

Perception matrix T_i of individual i is the transformation of an opinion x in objective coordinates into a subjective reference frame:

$$x = x'_1 \cdot \left(\sqrt{\lambda_1} \cdot \vec{v}_1 \right)_j + x'_2 \cdot \left(\sqrt{\lambda_2} \cdot \vec{v}_2 \right)_j + \dots =: T_i \cdot x' \quad (3)$$

$$\Longleftrightarrow \quad x' = (T_i)^{-1} \cdot x .$$

References



Enders, A. M., & Armaly, M. T. (2019). The Differential Effects of Actual and Perceived Polarization. *Political Behavior*, 41(3), 815–839. <https://doi.org/10.1007/s11109-018-9476-2>
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Levendusky, M. S., & Malhotra, N. (2016). (Mis)perceptions of Partisan Polarization in the American Public. *Public Opinion Quarterly*, 80(S1), 378–391. <https://doi.org/10.1093/poq/nfv045>
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Westfall, J., Van Boven, L., Chambers, J. R., & Judd, C. M. (2015). Perceiving Political Polarization in the United States: Party Identity Strength and Attitude Extremity Exacerbate the Perceived Partisan Divide. *Perspectives on Psychological Science*, 10(2), 145–158. <https://doi.org/10.1177/1745691615569849>